

Loops

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The Philosophy of Repetition



Why Do We Need Loops?

Scenario: You need to calculate the average GPA of 60 students in a CSE section.

Without Loops:

- `scanf("%f", &gpa1);`
- `scanf("%f", &gpa2);`
- ... (Write this 60 times?)
- `avg = (gpa1 + gpa2 + ... + gpa60) / 60;`

With Loops:

- Write the logic once.
- Tell the computer to repeat it 60 times.

Rule of thumb: If you are copy-pasting code more than twice, you likely need a loop.

Types of Loops in C

Loop	Control Type	Best Use Case
while	Entry-controlled	Unknown iterations
do-while	Exit-controlled	Run at least once
for	Entry-controlled	Known iterations

Key Insight: All three loops are interchangeable in functionality, but choosing the right one improves code readability

The Three Essential Components

Every loop requires three components to work correctly:

1. **Initialization:** Set the starting point (e.g., $i = 0$)
2. **Condition:** Define when to continue/stop (e.g., $i < 10$)
3. **Update:** Change the loop variable (e.g., $i++$)

Common Bug

Forgetting the **update step** leads to an **infinite loop**!

The while Loop

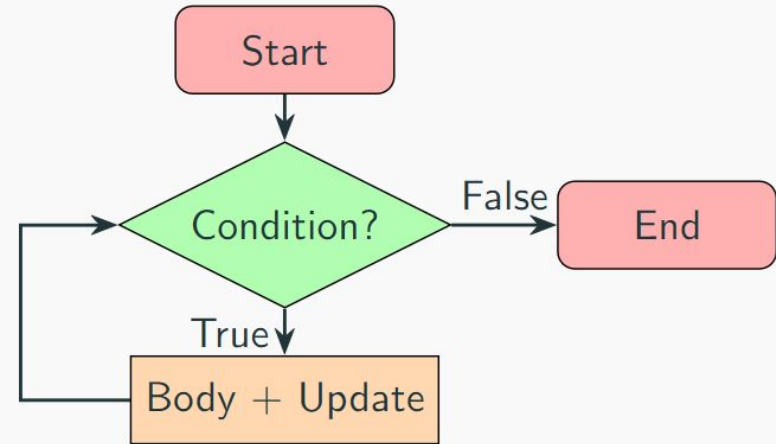


The while Loop: Syntax

The while loop checks the condition before executing the body.

Syntax

```
// Initialization BEFORE the loop  
int i = 0;  
  
while (condition) {  
    // Loop body  
    // Update step INSIDE  
    i++;  
}
```



Key Point: If condition is initially false, the loop body never runs.

Example: The Countdown

```
#include<stdio.h>
int main() {
    int count = 5; // Initialization

    while (count > 0) { // Condition
        printf("%d", count);
        count--; // Update - CRITICAL !
    }
    printf("Blastoff!\n");
    return 0;
}
```

Output: 5 4 3 2 1 Blastoff!

Trace:

- count=5: Print 5, decrement to 4
- count=4: Print 4, decrement to 3
- ... continues until count=0, condition fails

Think First: What Happens Here?

Question: What is the output of this code?

```
int x = 10;
while(x > 0) {
    printf("%d ", x);
    x = x - 3;
}
printf("Done");
```

Answer: 10 7 4 1 Done

Trace:

- x=10: Print 10, x becomes 7
- x=7: Print 7, x becomes 4
- x=4: Print 4, x becomes 1
- x=1: Print 1, x becomes -2
- x=-2: Condition fails, exit loop

Common Mistake: The Semicolon Trap

Question: What is the output of this code?

```
int i = 0;
while (i < 5); // <-- Notice the semicolon!
{
    printf("%d ", i);
    i++;
}
```

Answer: Infinite Loop! (No output printed)

Trace:

- The semicolon after `while(i < 5)` acts as an empty body.
- The program gets stuck repeatedly checking `0 < 5` and doing nothing.
- The block `{ . . . }` below is treated as separate, unreachable code.

Example: Sum of Digits

Problem: Find the sum of digits of a number (e.g., $1234 \rightarrow 1+2+3+4 = 10$)

```
int num = 1234, sum = 0;

while (num > 0) {
    int digit = num % 10; // Extract last digit
    sum = sum + digit; // Add to sum
    num = num / 10; // Remove last digit
}

printf("Sum of digits: %d\n", sum);
```

Trace:

- num=1234: digit=4, sum=4, num=123
- num=123: digit=3, sum=7, num=12
- num=12: digit=2, sum=9, num=1
- num=1: digit=1, sum=10, num=0
- num=0: Condition fails, output 10

Example: Count Digits in a Number

Problem: How many digits are in a given number?

```
int num, count = 0;
printf("Enter a number: ");
scanf("%d", &num);

// Handle special case of 0
if (num == 0) {
    count = 1;
} else {
    int temp = num;
    if (temp < 0) temp = -temp ; // Handle negative

    while (temp > 0) {
        count++;
        temp = temp / 10;
    }
}
printf("Number of digits: %d\n", count);
```

Example: Reverse a Number

Problem: Reverse the digits (e.g., 1234 → 4321)

```
int num = 1234, reversed = 0;

while (num > 0) {
    int digit = num % 10;
    reversed = reversed * 10 + digit;
    num = num / 10;
}

printf("Reversed: %d\n", reversed);
```

Think: What happens with num = 1200?

Answer: Output is 21 (leading zeros are dropped in integers!)

Example: Finding GCD (Euclidean Algorithm)

Problem: Find GCD of two numbers using Euclid's method.

Algorithm: $\text{gcd}(a, b) = \text{gcd}(b, a \bmod b)$ until $b = 0$

```
int a = 48, b = 18;

while (b != 0) {
    int temp = b;
    b = a % b;
    a = temp;
}

printf("GCD is %d\n", a);
```

Trace:

- $a=48, b=18$: $\text{temp}=18, b=48\%18=12, a=18$
- $a=18, b=12$: $\text{temp}=12, b=18\%12=6, a=12$
- $a=12, b=6$: $\text{temp}=6, b=12\%6=0, a=6$
- $b=0$: Exit. GCD = 6

The do-while Loop

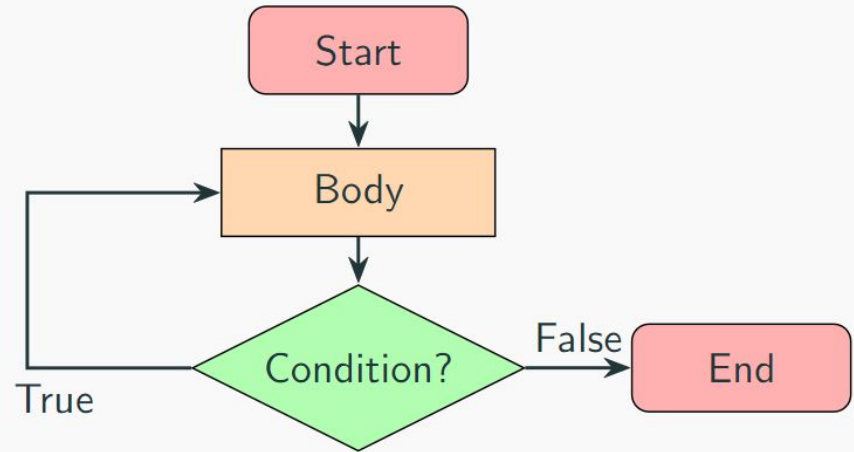


The do-while Loop: Exit Controlled

The body is executed at least once, regardless of the condition.

Syntax

```
do {  
  // Loop body  
  // ( includes update )  
} while (condition);  
// -- Semicolon required !
```



Don't forget the semicolon after while(condition);

Use Case: Input Validation

do-while is perfect when you need to ask the user for input until they get it right

```
int number;

do {
    printf("Enter a positive number (1 -100): ");
    scanf("%d", &number);

    if (number <= 0 || number > 100) {
        printf("Invalid! Try again.\n");
    }
} while (number <= 0 || number > 100);

printf("You entered: %d\n", number);
```

Key Point: We MUST ask at least once before we can check!

Use Case: Menu-Driven Program

```
int choice;

do {
    printf("\n=== Calculator ===\n");
    printf("1. Add\n2. Subtract\n3. Exit\n");
    printf("Enter choice: ") ;
    scanf("%d", &choice);

    switch (choice) {
        case 1: printf("Addition selected\n"); break;
        case 2: printf("Subtraction selected\n"); break;
        case 3: printf("Goodbye!\n"); break;
        default: printf("Invalid choice!\n");
    }
} while (choice != 3);
```

Think First: while vs do-while

Question: What is the output of each?

Code A:

```
int x = 0;
while (x > 0) {
    printf("%d", x);
    x--;
}
printf("End");
```

Code B:

```
int x = 0;
do {
    printf ("%d", x);
    x--;
} while (x > 0);
printf("End");
```

Answer A: End (body never executes, condition false initially)

Answer B: 0 End (body executes once, then condition checked)

Comparison: while vs do-while

Situation	Recommended Loop
Entry Controlled	Exit Controlled
Checks condition first	Executes body first
May run 0 times	Runs at least 1 time
No semicolon at end	Ends with semicolon
Use when iteration might be 0	Use for menus, input validation

Mental Model:

- `while`: "Check, then do" (might not do anything)
- `do-while`: "Do, then check" (always does at least once)

The for Loop



The for Loop: Anatomy

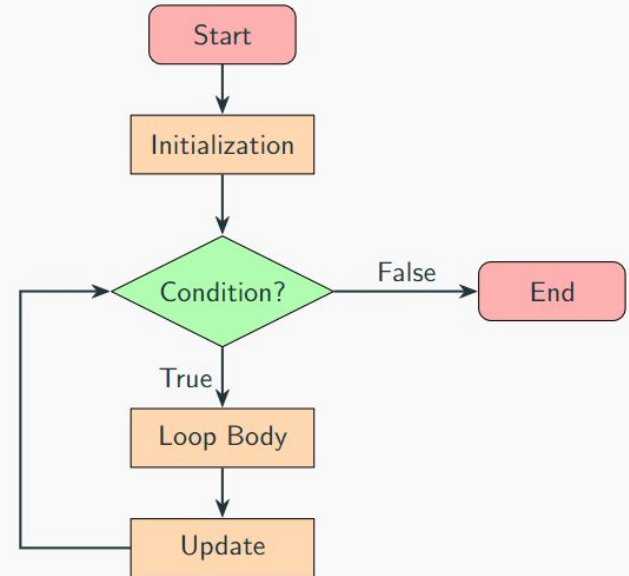
Best used when the **number of iterations is known** beforehand.

Syntax

```
for ( init ; condition ; update )  
{  
    // loop body  
}
```

Execution Order:

1. **Initialization:** Runs once at the start (e.g., $i = 0$)
2. **Condition:** Checked before each iteration (e.g., $i < 10$)
3. **Body:** Executes if condition is true
4. **Update:** Runs after the body (e.g., $i++$)
5. Go to step 2



Example: Print Numbers 1 to N

```
int n;  
printf("Enter N: ");  
scanf("%d", &n);  
  
for(int i = 1; i <= n; i++) {  
    printf("%d", i);  
}  
printf("\n");
```

For N=5: Output is 1 2 3 4 5

Trace:

- $i=1$: $1 \leq 5$? Yes. Print 1. i becomes 2.
- $i=2$: $2 \leq 5$? Yes. Print 2. i becomes 3.
- ... continues ...
- $i=6$: $6 \leq 5$? No. Exit loop.

Example: Sum of Series

Calculate $S = 1 + 2 + 3 + \dots + N$

```
int n, sum = 0;
printf("Enter N: ");
scanf("%d", &n);

for (int i = 1; i <= n ; i++) {
    sum = sum + i;
}

printf("Sum = %d\n", sum);
```

Example: Factorial Calculation

Calculate $N! = 1 \times 2 \times 3 \times \dots \times N$

```
int n ;  
long long factorial = 1; // Use long long for big numbers  
  
printf("Enter N: ");  
scanf("%d", &n);  
  
for (int i = 1; i <= n ; i++) {  
    factorial = factorial * i;  
}  
  
printf("%d! = %lld\n", n, factorial);
```

Note: Factorial grows very fast. 13! exceeds int range

Variations: Counting Down

```
// Print 10 to 1  
for (int i = 10; i >= 1; i--) {  
    printf("%d", i);  
}  
// Output : 10 9 8 7 6 5 4 3 2 1
```

```
// Print even numbers from 2 to 20  
for (int i = 2; i <= 20; i += 2) {  
    printf("%d", i);  
}  
// Output : 2 4 6 8 10 12 14 16 18 20
```

```
// Powers of 2: 1 , 2 , 4 , 8 , ...  
for (int i = 1; i <= 1024; i *= 2) {  
    printf("%d", i);  
}  
// Output : 1 2 4 8 16 32 64 128 256 512 1024
```

Advanced: Multiple Variables

You can have multiple variables in a for loop!

```
// Two variables moving towards each other  
for (int i = 0, j = 10; i < j; i++, j--) {  
    printf("i =%d, j =%d\n", i, j);  
}
```

Output:

```
i=0, j=10  
i=1, j=9  
i=2, j=8  
i=3, j=7  
i=4, j=6
```

Next iteration: i=5, j=5. Condition $i < j$ fails (5 < 5 is false)

Optional Parts: Infinite Loop

All three parts of the for loop header are optional!

```
// Infinite loop - all parts empty
for ( ; ; ) {
    printf("Forever...\n");
    // Use break to exit , or Ctrl + C
}
```

This is equivalent to `while(1) { ... }`

```
// Initialization outside, update inside
int i = 0;
for ( ; i < 5 ; ) {
    printf ("%d", i);
    i++;
}
```

Think First

Question 1: What is the output?

```
for (int i = 0; i < 5; i++);  
    printf("%d", i);
```

Answer: Compilation ERROR! Variable `i` is not visible outside the loop.

Question 2: What about this?

```
int i;  
for(i = 0; i < 5; i++);  
    printf("%d", i);
```

Answer: Prints 5. The semicolon makes an empty loop body. Loop runs with `i=0,1,2,3,4`. When `i=5`, condition fails. Then we print `i`.

Critical: The Floating Point Trap

Never use floating point numbers as loop counters for exact equality!

```
for (float x = 0.0; x != 1.0; x += 0.1) {  
    printf("%f ", x);  
}
```

Why is this dangerous?

- Computers cannot represent 0.1 exactly in binary
- Adding 0.1 ten times might give 0.99999... or 1.00001
- The condition $x \neq 1.0$ may remain true forever!

Safe Alternative:

```
for ( float x = 0.0; x < 1.0; x += 0.1) { ... }  
// OR use integer counter :  
for ( int i = 0; i < 10; i++) { float x = i * 0.1; ... }
```

Nested Loops



The Concept of Nesting

Analogy: A Clock

- The Second hand (Inner Loop) must make a full circle (60 ticks)...
- ..for the Minute hand (Outer Loop) to move just once.
- The Minute hand makes 60 ticks for the Hour hand to move once.

In Programming:

- The inner loop completes ALL its iterations for EACH iteration of the outer loop.
- Outer loop often controls Rows
- Inner loop often controls Columns

Total iterations: Outer count $\times$ Inner count

Example: Printing a Rectangle

```
for (int row = 1; row <= 3; row++) {  
    for (int col = 1; col <= 5; col++) {  
        printf ("* ");  
    }  
    printf("\n"); // Newline after each row  
}
```

Output:

* * * * *

* * * * *

* * * * *

Total stars: $3 \times 5 = 15$

Example: Multiplication Table

```
for (int i = 1; i <= 5; i++) {  
    for (int j = 1; j <= 5; j++) {  
        printf("%4d", i * j); // %4d for alignment  
    }  
    printf("\n");  
}
```

Output:

1	2	3	4	5
2	4	6	8	10
3	6	9	12	15
4	8	12	16	20
5	10	15	20	25

Dependent Inner Loops: Right Triangle

What if the inner loop depends on the outer loop variable?

```
for (int i = 1; i <= 5; i++) {  
    for (int j = 1; j <= i; j++) { // j goes up to i!  
        printf("* ");  
    }  
    printf("\n");  
}
```

Output:

```
*  
* *  
* * *  
* * * *  
* * * * *
```

Pattern: Number Triangle

```
for (int i = 1; i <= 5; i++) {  
    for (int j = 1; j <= i ; j++) {  
        printf("%d", j);  
    }  
    printf("\n");  
}
```

Output:

```
1  
1 2  
1 2 3  
1 2 3 4  
1 2 3 4 5
```

Think First: Inverted Triangle

Question: What code produces this output?

```
* * * * *  
* * * *  
* * *  
* *  
*
```

Answer:

```
for (int i = 5; i >= 1; i--) {  
    for (int j = 1; j <= i ; j++) {  
        printf("* ");  
    }  
    printf("\n");  
}
```

Pyramid Pattern

Question: Print a centered pyramid

```
  *
 ***
*****
*****
*****
```

Analysis for row i (1 to n):

- Spaces before stars: $n - i$
- Number of stars: $2i - 1$

Pyramid Solution

```
int n = 5;
for (int i = 1; i <= n ; i++) {
    // Print spaces
    for (int s = 1; s <= n - i; s++) {
        printf(" ");
    }
    // Print stars
    for (int j = 1; j <= 2* i - 1; j++) {
        printf("* ");
    }
    printf("\n");
}
```


Question	My Answer
CSE 273 1. Write a program to print the pattern. <pre>1 4 6 9 11 13 17 19 21 23</pre>	<pre>#include <stdio.h> int main() { printf("1\n"); printf("4 6\n"); printf("9 11 13\n"); printf("17 19 21 23"); return 0; }</pre> 

Moral: This is why we learn loops!
(Hardcoding works... until the pattern changes to 100 rows)

Jump Statements: break & continue



Control Flow Modifiers

Sometimes we need to alter the normal flow of a loop.

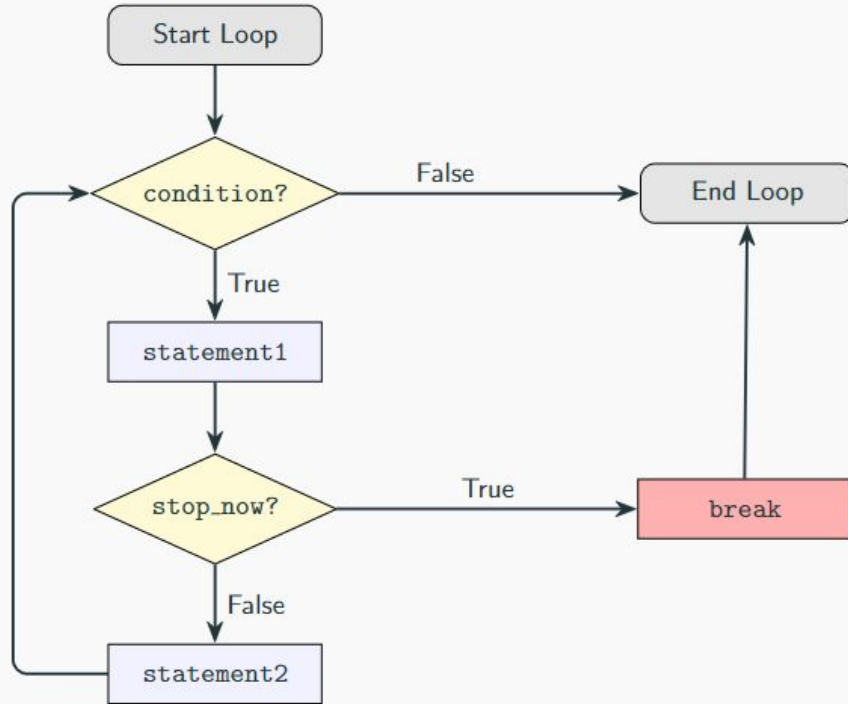
break Terminates the loop **immediately**. Control jumps to the statement after the loop body.

continue Skips the **rest of the current iteration**. Control jumps to:

- `for`: the update expression
- `while/do-while`: the condition check

Important: Both only affect the **innermost** loop they're in!

Flowchart: The break Statement



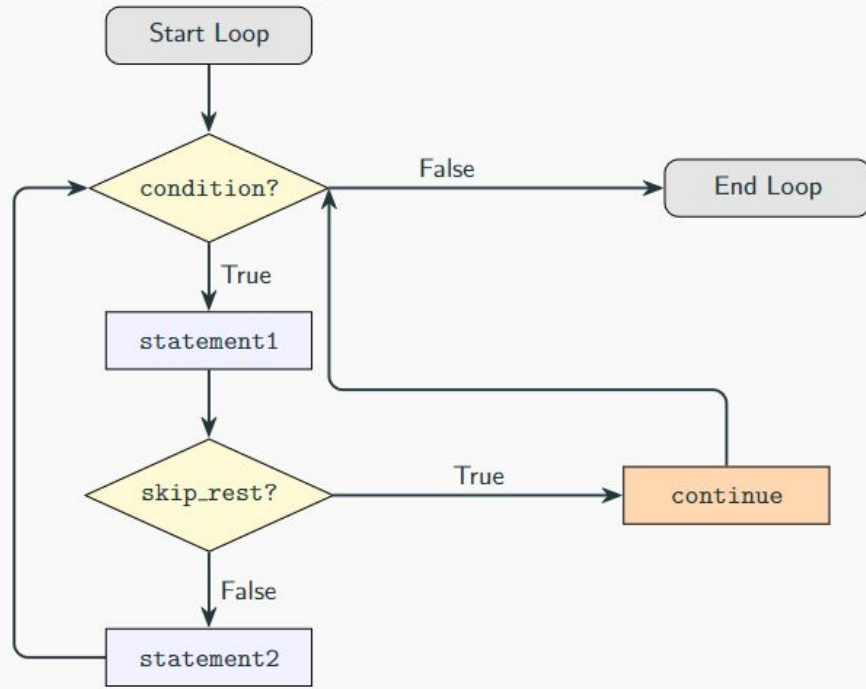
Behavior:

- Terminates the loop immediately.
- Moves control to the statement just after the loop.

Example Logic:

```
while (condition) {  
    statement1;  
    if (stop_now) {  
        break; // Exit loop  
    }  
    statement2;  
}  
// Resumes here
```

Flowchart: The continue Statement



Behavior:

- Skips the rest of the current iteration (Part 2).
- Jumps back to evaluate the condition (or update).

Example Logic:

```
while (condition) {  
    statement1;  
    if (skip_rest) {  
        continue; // Next iter  
    }  
    statement2; // Skipped  
}
```

Example: break - Finding First Divisor

```
int n = 100;
printf("First divisor of %d (other than 1): ", n);

for (int i = 2; i <= n ; i++) {
    if (n % i == 0) {
        printf("%d\n", i);
        break; // Found it , stop searching!
    }
}
```

Output: First divisor of 100 (other than 1): 2

Without break: Would print ALL divisors (2, 4, 5, 10, 20, 25, 50, 100)

Example: break - Prime Checker

```
int n = 29, isPrime = 1;

for (int i = 2; i * i <= n ; i++) { // Only check up to sqrt(n)
    if (n % i == 0) {
        isPrime = 0;
        break; // Found a factor , no need to check more
    }
}

if (isPrime && n > 1)
    printf("%d is Prime\n", n);
else
    printf("%d is not Prime\n", n);
```

Efficiency: For $n=1000000$, we only check up to 1000 instead of 999999!

Example: continue - Skip Specific Values

Print 1 to 10, but skip multiples of 3:

```
for (int i = 1; i <= 10; i++) {  
    if (i % 3 == 0) {  
        continue; // Skip rest of body, go to i++  
    }  
    printf("%d ", i);  
}
```

Output: 1 2 4 5 7 8 10

(Skipped: 3, 6, 9)

Example: Sum of Non-Negative Numbers

Sum numbers until user enters negative (sentinel value):

```
int num, sum = 0;

while (1) { // Infinite loop
    printf("Enter a number (negative to stop): ");
    scanf("%d", &num);

    if (num < 0) {
        break; // Exit the loop
    }
    sum += num;
}

printf("Sum = %d\n", sum);
```

Danger: continue in while Loop

Be careful with continue in while loops!

```
int i = 0;
while (i < 5) {
    if (i == 2) {
        continue; // BUG: i is never incremented!
    }
    printf("%d", i);
    i++;
}
```

Result: Infinite loop! When $i=2$, continue skips $i++$.

Fix:

```
int i = 0;
while (i < 5) {
    if (i == 2) { i++; continue; } // Increment first!
    printf("%d ", i);
    i++;
}
```

Nested Loops: break Only Exits Innermost

```
for (int i = 1; i <= 3; i++) {  
    printf("Outer %d: ", i);  
    for (int j = 1; j <= 5; j++) {  
        if (j == 3) break; // Only breaks inner loop  
        printf("%d ", j);  
    }  
    printf("\n");  
}
```

Output:

Outer 1: 1 2

Outer 2: 1 2

Outer 3: 1 2

The outer loop continues! Each inner loop stops at j=3.

Breaking Out of Multiple Loops

Problem: How to exit all loops at once?

Solution 1: Use a flag variable

```
int found = 0;
for (int i = 0; i < 10 && !found; i++) {
    for (int j = 0; j < 10 && !found; j++) {
        if (arr[i][j] == target) {
            printf ("Found at (%d, %d)\n", i, j);
            found = 1; // This will stop both loops
        }
    }
}
```

Solution 2: Use goto (not recommended in general)

Some Random Questions



The Unsigned Trap

What happens with this countdown?

```
unsigned int i;  
for (i = 10; i >= 0; i--) {  
    printf("%u ", i);  
}
```

Infinite Loop!

- unsigned int can NEVER be negative
- When i is 0 and we do i--, it wraps to UINT MAX (≈ 4 billion)
- Condition $i \geq 0$ is ALWAYS true for unsigned

Fix:

```
for (unsigned int i = 10; i > 0; i--) { printf ("%u ", i); }  
printf("0 "); // Print 0 separately  
// OR use signed int
```

Pre vs Post Increment

```
int i = 0;
while (i++ < 3) {
    printf("%d ", i);
}
printf("Final: %d", i);
```

Output: 1 2 3 Final: 4

Trace:

1. Check $(0 < 3) = \text{True}$, THEN i becomes 1. Print 1.
2. Check $(1 < 3) = \text{True}$, THEN i becomes 2. Print 2.
3. Check $(2 < 3) = \text{True}$, THEN i becomes 3. Print 3.
4. Check $(3 < 3) = \text{False}$, THEN i becomes 4. Exit.

Empty Loop Bodies

What does this compute?

```
int n = 12345, count = 0;  
while (n > 0) n /= 10, count++;  
printf("%d", count);
```

Answer: 5 (counts number of digits)

The comma combines both statements into the condition area!

Another one:

```
int s = 0, i = 1;  
while (i <= 100) s += i++;  
printf("%d", s);
```

Answer: 5050 (sum of 1 to 100)

Problem: Prime Numbers in Range

Print all prime numbers between 2 and 50:

```
for (int n = 2; n <= 50; n++) {  
    int isPrime = 1;  
  
    for (int i = 2; i * i <= n; i++) {  
        if (n % i == 0) {  
            isPrime = 0;  
            break;  
        }  
    }  
  
    if (isPrime) {  
        printf("%d ", n);  
    }  
}
```

Output: 2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 50

Problem: Fibonacci Series

Print first N Fibonacci numbers: 0, 1, 1, 2, 3, 5, 8, 13, ...

```
int n = 10;
int a = 0, b = 1;

printf ("%d %d ", a, b);

for (int i = 3; i <= n; i++) {
    int next = a + b;
    printf("%d ", next);
    a = b;
    b = next;
}
```

Output: 0 1 1 2 3 5 8 13 21 34

Problem: Diamond Pattern

Goal: Print a diamond with $n=5$

```
  *
 ***
*****
*****
*****
*****
*****
 ***
  *

```

Hint: Combine pyramid (top half) with inverted pyramid (bottom half)

Summary & Best Practices



Loop Selection Guide

Situation	Recommended Loop
Known number of iterations	for
Unknown iterations, might be 0	while
Must execute at least once	do-while
Processing arrays	for
Reading until sentinel value	while with break
Menu-driven programs	do-while
Searching/finding first match	for/while with break

Common Mistakes to Avoid

1. **Infinite loops:** Forgetting update step
2. **Off-by-one errors:** $<$ vs $<=$
3. **Semicolon after condition:** `while(x);` or `for(...);`
4. **Floating point counters:** Never use for exact comparison
5. **Unsigned underflow:** `unsigned i; i >= 0` always true
6. **continue in while:** May skip increment
7. **Breaking wrong loop:** `break` only exits innermost
8. **Uninitialized variables:** Always initialize loop counters

Best Practices

1. **Initialize all variables** before using them
2. **Use meaningful names:** `row`, `col` instead of `i`, `j` when appropriate
3. **Keep loop bodies simple:** Extract complex logic into functions later
4. **Use braces always**, even for single statements
5. **Limit nesting depth:** More than 3 levels is hard to read
6. **Comment complex loops:** Explain what pattern is being generated
7. **Test boundary conditions:** 0 iterations, 1 iteration, max value

Summary

1. **while:** Entry-controlled. Use when iterations unknown.
2. **do-while:** Exit-controlled. Guarantees at least one execution.
3. **for:** Best for counting. All parts (init, cond, update) in header.
4. **Nested loops:** Inner completes fully for each outer iteration. Think rows/columns.
5. **break:** Exit loop immediately.
6. **continue:** Skip to next iteration.

Acknowledgement

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